COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 6 : SLO – 2 | Binary Arithmetic – Multiplication and Division
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Section A – Match the Following (Answers)

- 1. Binary Multiplication → B. Use 0/1 multiplication and shifting
- 2. Binary Division → A. Repeated subtraction and shifting
- 3. Multiplying by 0 → C. Always gives 0
- 4. Multiplying by 1 → D. Gives the same number
- 5. Shift Left → E. Equivalent to multiplying by 2

Section B – Fill in the Blanks

- Binary multiplication uses only two operations: AND and SHIFT LEFT.
- Division in binary uses repeated SUBTRACTION with right shifts.
- Shifting left by one bit is equal to MULTIPLYING by 2.
- Shifting right by one bit is equal to DIVIDING by 2.
- If divisor is greater than the dividend, the quotient is 0 and the remainder is the DIVIDEND.

Section C – Activity Task (Binary Multiplication)

- (a) 1011 × 0101: partial products 1011 (×1), 0000 (×0), 1011 shifted 2 (×1), 0000 shifted 3 (×0); sum = 0110111. Answer: 1011 × 0101 = 0110111 (= 11 × 5 = 55).
- (b) 1101 × 0011: partial products 1101 (×1), 1101 shifted 1 (×1), 0000, 0000; sum = 100111. Note: 1101 × 0011 = 13 × 3 = 39 = 0100111; adding shifted partial products 1101 + 11010 = 100111 → Answer: 1101 × 0011 = 100111.

Section D – Activity Task (Binary Division)

- (a) 10110 ÷ 101: 101 goes into 1011 once (quotient bit 1), remainder 0110 brings down; 101 > 011 → quotient bit 0; final: Quotient = 100, Remainder = 10. Check: 10110■ = 22, 101■ = 5 → 22 = 5×4 + 2, so Quotient = 100 (4), Remainder = 010 (2).
- (b) 11001 ÷ 11: 11 into 110 twice (11), 11001 = 25, 11■ = 3 → 25 = 3×8 + 1, so Quotient = 1000 (8), Remainder = 1. Long division: 11 into 110 → 1 (rem 0), bring 0 → 00 < 11 → 0, bring 1 → 001 < 11 → 0, final remainder 1 → Quotient = 1000, Remainder = 001.

Section E – True or False

- Binary multiplication uses only 0, 1 and shifting. — TRUE
- Division by shifting right is exact for any binary number. — FALSE (exact only when no 1-bits are shifted out, e.g. even numbers)
- In binary multiplication, multiplying by 1 leaves the number unchanged. — TRUE
- In binary division, if the divisor is bigger than the partial dividend, a 0 is placed in the quotient. — TRUE
- Shifting left by two positions multiplies the number by 4. — TRUE